

Question number	Answer	Additional guidance	Mark
3(a)	An explanation that includes three of the following points: - the myelin sheath is an insulator (1) - therefore the action potential only occurs at nodes of Ranvier (1) - and therefore the nerve impulse 'jumps' from one node to the next which is known as saltatory conduction (1)	Accept The influx of sodium ions at one node creates enough depolarization to reach the threshold of the next.	(3)

Question number	Answer	Additional guidance	Mark
3(b)	An answer that includes two of the following points: - when the myelin sheath is destroyed transmission of the nerve impulse is impaired (1) - fewer nerve impulses reach the muscle (1) - fewer muscle fibres are stimulated to contract which leads to muscle weakness (1)	slower transmission of impulses	(2)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes two of the following points: • all types of dementia have a lower level of myelin than the control group (1) • there is no relationship between the amount of myelin in the brain and the type of dementia (1) • no overlap in any of the standard deviation for any type of dementia and control which suggests that the conclusions may be valid (1)		(2)